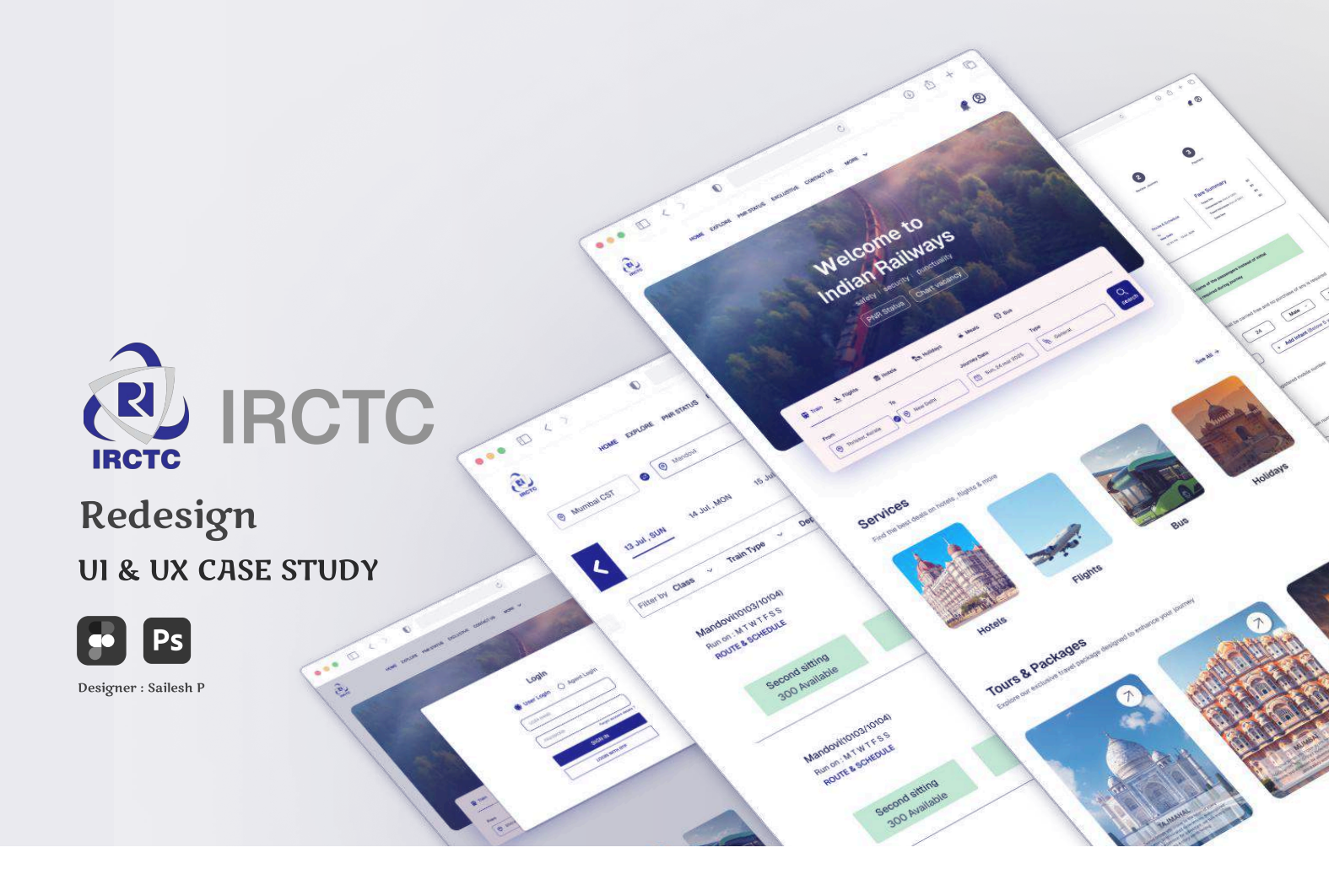

IRCTC
Redesign
UI & UX CASE STUDY
Ps
Designer : Sailesh P
Welcome to Indian Railways
Services
Hotels
Flights
Bus
Holidays
Tours & Packages
Login
Second sitting 300 Available

PROJECT OVERVIEW

Project Description:

Indian Railway Catering And Tourism Corporation (IRCTC) Is An Indian Railway Subsidiary Established By The Ministry Of Railways To Manage Indian Railway Catering, Online Ticketing And Tourism Operations.

Project Goal:

The Goal Of The Project Was To Redesign The Layout, Hierarchy, Flow, And Navigation And Make The Design Intuitive And Easy To Use According To The Latest Trends.

Role
Sole UI/UX Designer

Timeline
2 Month

Responsibities
UX/UI Researcher

Problem Statement

- Inconsistency, Bad Layout, Lack Of Hierarchy, Random Placement Of CTAS.
- Advertisements Were Very Close To The CTAs, Causing Users Trouble Finding And Using The CTAS.
- Users Were Unable To Select The Seats And Berths According To Their Preferences
- Complex And Lengthy Registration Process.
- Cluttered And Outdated Design Without Hierarchy Creates Confusion And Prevents Users From Reaching Useful Actions And Information.
- Captcha On Every Single Page Frustrated Users And Slowed Down The Website's Performance.
- Bad Navigation And Lengthy Process Make The Product Complex, And The More Complex It Is, The More Time Users Take To Make Decisions, Which Causes Cognitive Load.
- There Was No Option To Get The Running Status Of The Train.

Solution Statement

- Provided Proper Layout, Hierarchy, And Fixed Place For CTAS
- Changed The Placement Of The Advertisements So That They Do Not Cause Any Trouble To The Users.
- Added The Drop-Down Menu From Which Users Can Select Seats And Berths According To Their Preferences.
- Provided Simple And Short Registration Process
- Provided Clean And Minimal Design With A Proper Hierarchy, Which Allows Users To Take Actions Or Scan The Information Easily Without Any Confusion
- Only One Captcha On The Login Page To Differentiate Between Humans And Computers For Security Purposes
- Designed A Simple User Flow And Short Process To Complete The Task, Which Takes Users Less Time, Makes Them Take Decisions Fast, And Reduces The Cognitive Load.
- Added The Option To Check The Running Status Of The Train.

Design Process

Discover → Define → Ideate → Design → Test

Why I Redesign IRCTC?

I've Been Using IRCTC To Book Train Tickets For Many Years, And During That Time, I've Encountered Numerous Issues With This Website, Such As

- Unable To Log In Due To The Perplexing Captcha Design That Is Stuck With The Advertisement
- It Was Necessary To Log In Even To Search For The Trains, And I Had To Enter My Username And Password To Search For Trains.
- There Was A Captcha On Every Single Page, Which Was Time-Consuming And Irritating.
- The User Interface Was Cluttered And Confusing.
- Booking A Ticket With The IRCTC Was Like Wasting 10 To 15 Minutes Because Of That Bad Design, So I Decided To Redesign It.

Above Are The Only Few Pain Points That I Describe; There Are A Lot More. I Was Aware Of The Bad Design Of The IRCTC Website, But I Wanted To Know What Other Users Were Feeling Or Saying About It, Which Is Why I First Decided To Read The User Reviews, And For That, I Visited The MouthShut Website.

Discover Phase

Qualitative Research

After reading a lot of reviews, I decided to conduct interviews to learn about the experiences of locals with the IRCTC website. So I interviewed some of my friends and family members over a phone call to know the pain points or challenges they face while booking tickets through the ICTC website.

A few of the questions I asked in an interview are listed below.

- Please describe any issues you recently encountered while booking tickets through IRCTC
- Please explain your best experience with the IRCTC website
- Please explain your worst experience with the IRCTC website.
- Please explain what can make your experience better with IRCTC or what should be improved on the website so that you can book tickets easily.
- If you could change one thing about the design, what would it be and why?

Insights Derived From The Interview

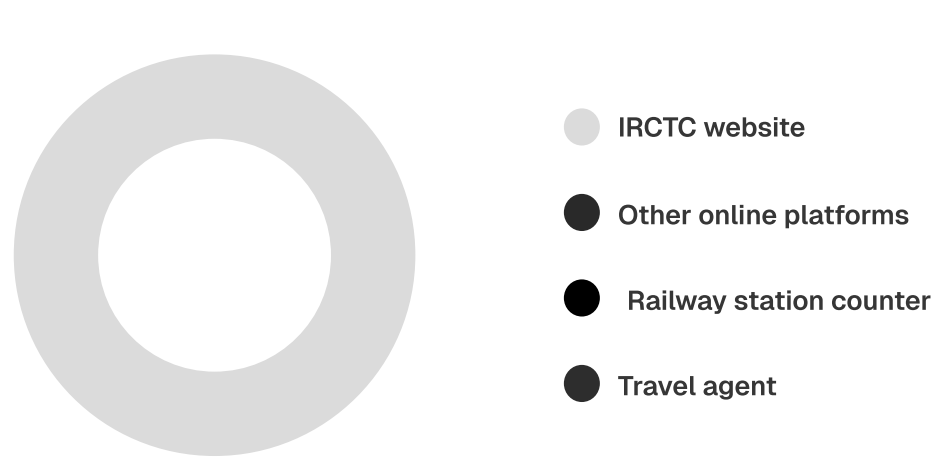
- Automatic Logout Is Frequent And Sometimes It Is Very Frustrating
- Too Much Information On The Booking Page Is Hard To Observe At Once.
- Users Have To Click On The Refresh Button To See The Seat Availability.
- Users Are Frustrated By Advertisements Because They Pop Up Right In The Middle Of The Process.
- For A Lot Of Users, Navigation Is Confusing
- Automatic Log Out Time Should Be Increased
- Elderly Users Cannot Find The Print Option For The Ticket.
- Users Want Clean Login Page.

Quantitative Research

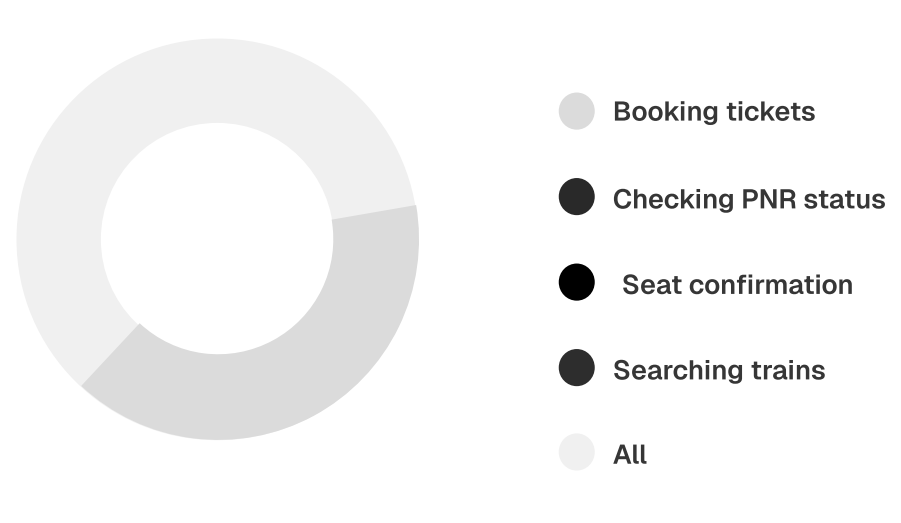
I did an online survey using Google Forms to find out the common problems and needs of the potential users. A total of 20 people responded, which enabled me to properly outline the issue.

Questions that I asked in a survey are below.

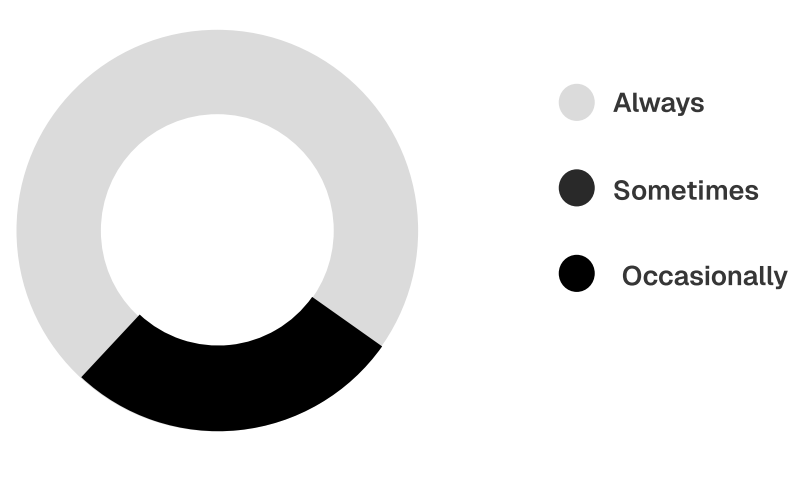
How do you book train tickets?



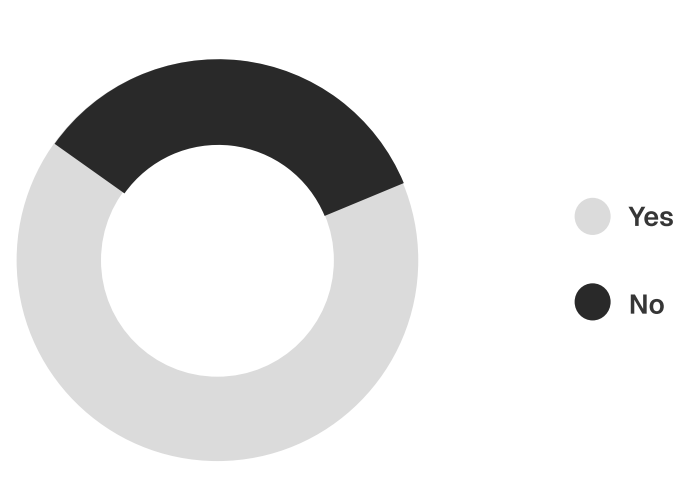
What do you use the IRCTC website for?



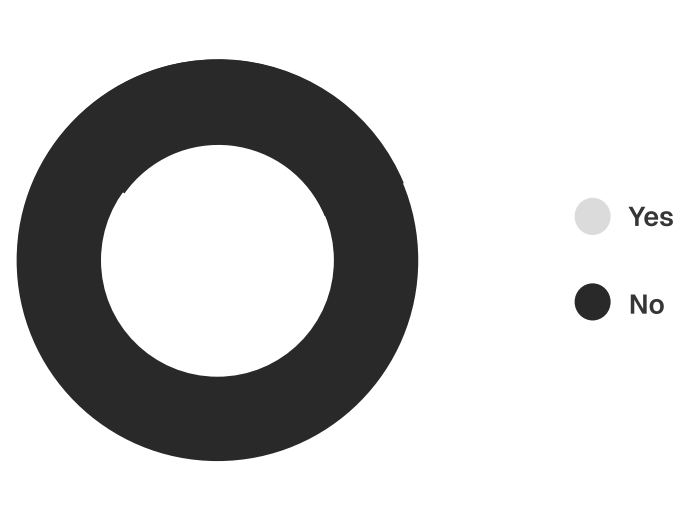
How often do you book tickets from the IRCTC?



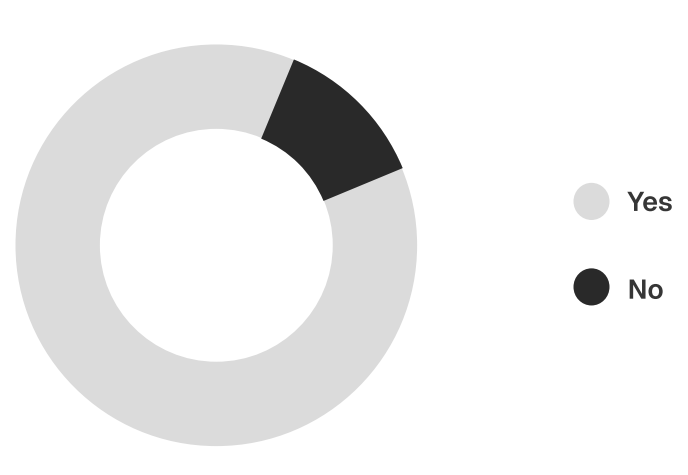
Is IRCTC website easy to use?



Do you face only problem finding the history of your tickets



Can ypu cancel your ticket easily ?



Insights Derived From The Survey

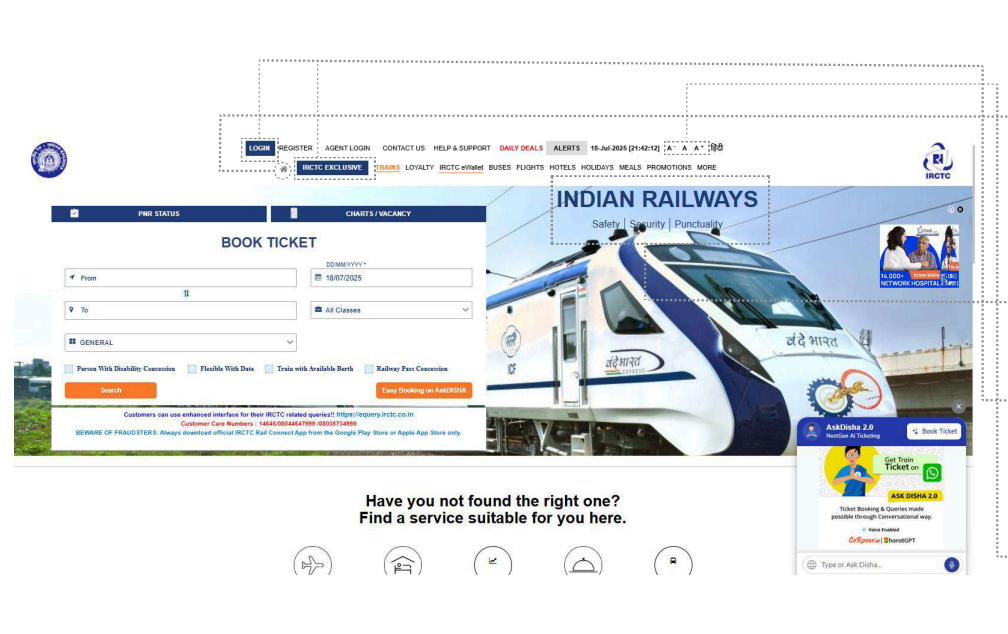
- Although IRCTC Has A Bad Design, Most Users Still Use It Instead Of Other Online Or Offline Services
- Both Advertisements And Captcha Codes Create Problems For Users.
- IRCTC's Website Is Confusing For New Users, But Regular Users Are Now Used To Its Bad Design.
- Users Are Frustrated By Advertisements Because They Pop Up Right In The Middle Of The Process

Following The Research, I Had A Good Idea Of What Users' Pain Points Are And What They Expect From The Design For A Better Experience. So Next, I Decided To Do Competitive Analysis And SWOT Analysis To Know What Competitors Are Offering, What Makes Their Design Better Than IRCTC, What Are Their Strengths And Weaknesses, And What Are The Areas Where They Need Improvement

UX Audit

Landing Page

Navigation Bar Of The Website Is Cluttered And Overloaded With Information


BOOK TICKET
INDIAN RAILWAYS
Have you not found the right one?
Find a service suitable for you here.

Home Icon
This Home Icon Is Very Small; Users Will Not Even Notice It At First Sight, And It Is Not Looking Good Because It Is The Only Icon Between The Text. They Should Have Used The Text Instead

Banner Heading
Text Is Not Visible.

CTA
Login Is A Button That Will Lead To The Log-In Page. While IRCTC Exclusive Is A Drop-Down Menu Then Why Is It Designed As A Button? It Cannot Look The Same As The Login Button

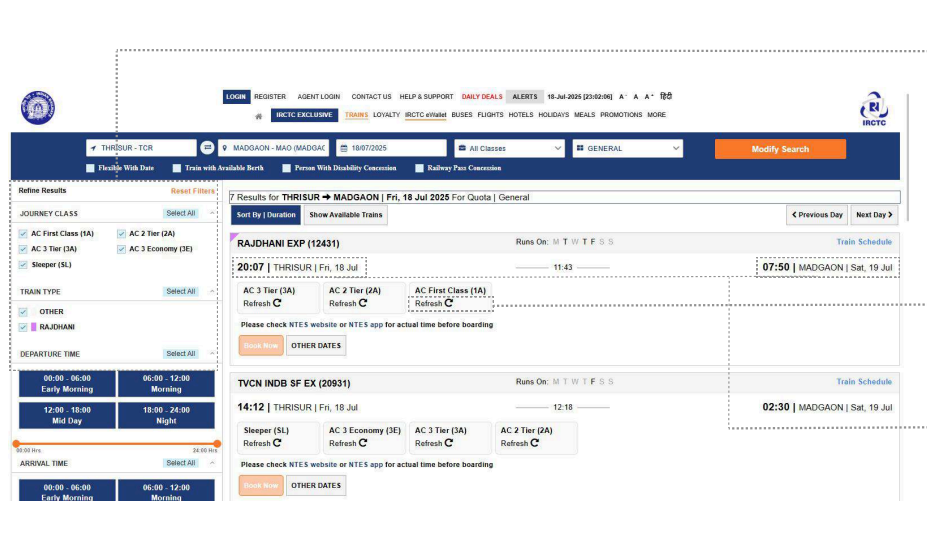
A-, A, A+
Text Resizing Options (A-, A, A+) Are Available But Poorly Placed And Not Intuitive. Also, The Color Contrast For Certain Sections Does Not Follow Accessibility Guidelines.

Lack Of Visual Hierarchy
There's Minimal Contrast Or Spacing Between Primary And Secondary Actions. The "Book Ticket" Section Competes For Attention With Promotional Content And Pop-Ups Like "Ask Disha" Chat.

Cluttered Interface
The Homepage Is Overloaded With Features, Options, Banners, And Advertisements, Which Overwhelms Users—Especially First-Time Visitors. Important Tasks Like Booking A Ticket Are Buried Under Unnecessary Visual Noise.

Search Result / Train List

The Page That Shows The Train List Looks Horrible Because Of The Bad Hierarchy, Cluttered Design, Misleading CTAs, Lack Of Consistency, And More



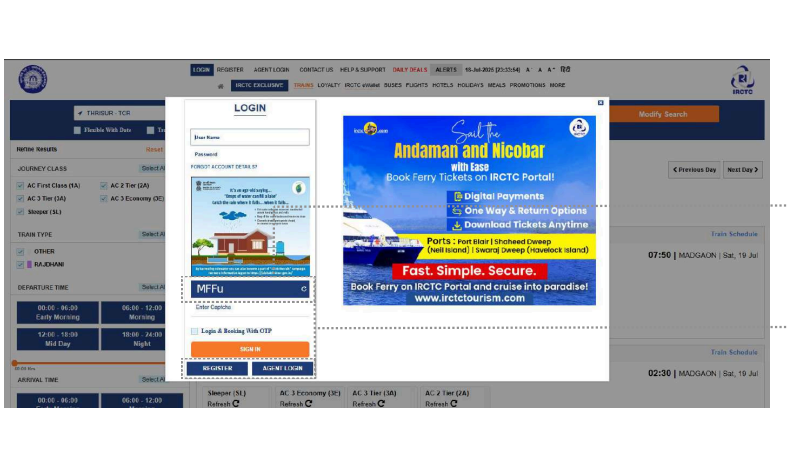
Filter
Users Have To Scroll All The Way Down To See The Full Filter. It Can Be Improved Because There Is A Lot Of Information To Absorb At Once.

Seat Availability
Users Have To Click On Refresh To See Available Seats

Station Information
Users Have To Look All The Way From Left To Right To See The Information About Boarding And Destination Stations

Poor Visual Hierarchy In Search Results
Train Names, Timings, And Class Information Are Crammed Together With Minimal Spacing Or Contrast. Primary CTAs Like "Book Now" Are Small And Inactive (In Grey), Even When They Could Provide Useful Real-Time Feedback Or Alternative Suggestions.

Login


LOGIN
Andaman and Nicobar
Fast. Simple. Secure.

Captcha
The Captcha Code Is Stuck To The Advertisement Which Could Cause Confusion For Users If It Has Same-Colored Advertisements

Confusing Design
Again, They Confuse Users By Desigriing Captchas Exactly The Same As CTAS (Buttons).

Unbalanced Layout
The Split Design (Form On The Left, Ad On The Right) Creates An Imbalanced Focus. The Promotional Section Is Brighter, More Colorful, And Visually Dominant—Stealing Attention From The Login Form.

Poor Form Hierarchy And Alignment
The Login Form Fields (Username, Password, Captcha) Lack Proper Spacing And Alignment. Important Buttons Like Sign In And Register Are Not Visually Prioritized, And The Checkbox ("Login & Booking With OTP") Looks Cramped.

User Persona

A Persona In User-Centered Design Is A Fictional Character To Representa User Type That Might Use A Site Brand Or Product In A Similar Way

Riya R.V
Ux Researcher

AGE : 27
GENDER : FEMALE
EDUTACATION : MCA
LOCATION : USA

Bio

Riya R.V Is A Researcher Who Loves To Travel And Explore Nature. She Loves Travelling To Different Places In Her Bike Usually On Weekends. She Feels Frustrated Whe She Has To Visit A New Place As She Cant Find A Single Source Which Gives Her Complete Information About Her Travle Destination.

Goals
- Search The Mans And Book Ticket Easily Without Visiting The Railway Ticket
- Get A Lower Berth For Long Journeys
- Download Tickets Easily

Frustrations
- Problem In Registering An Account Because Of The Length Process
- Confusion Due To A Lot Of Advertisment
- Trouble In Navigation

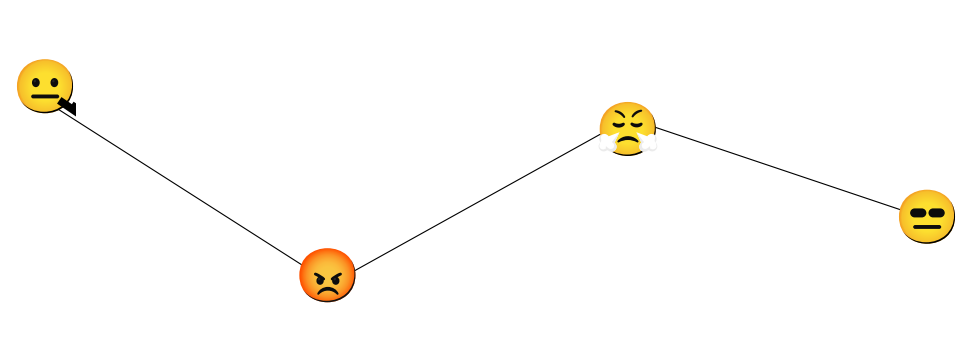
Technology
- Social Network
- Android
- Travel Apps
- Laptop

Motivations
- Food
- Incentive
- Fides

User Journey

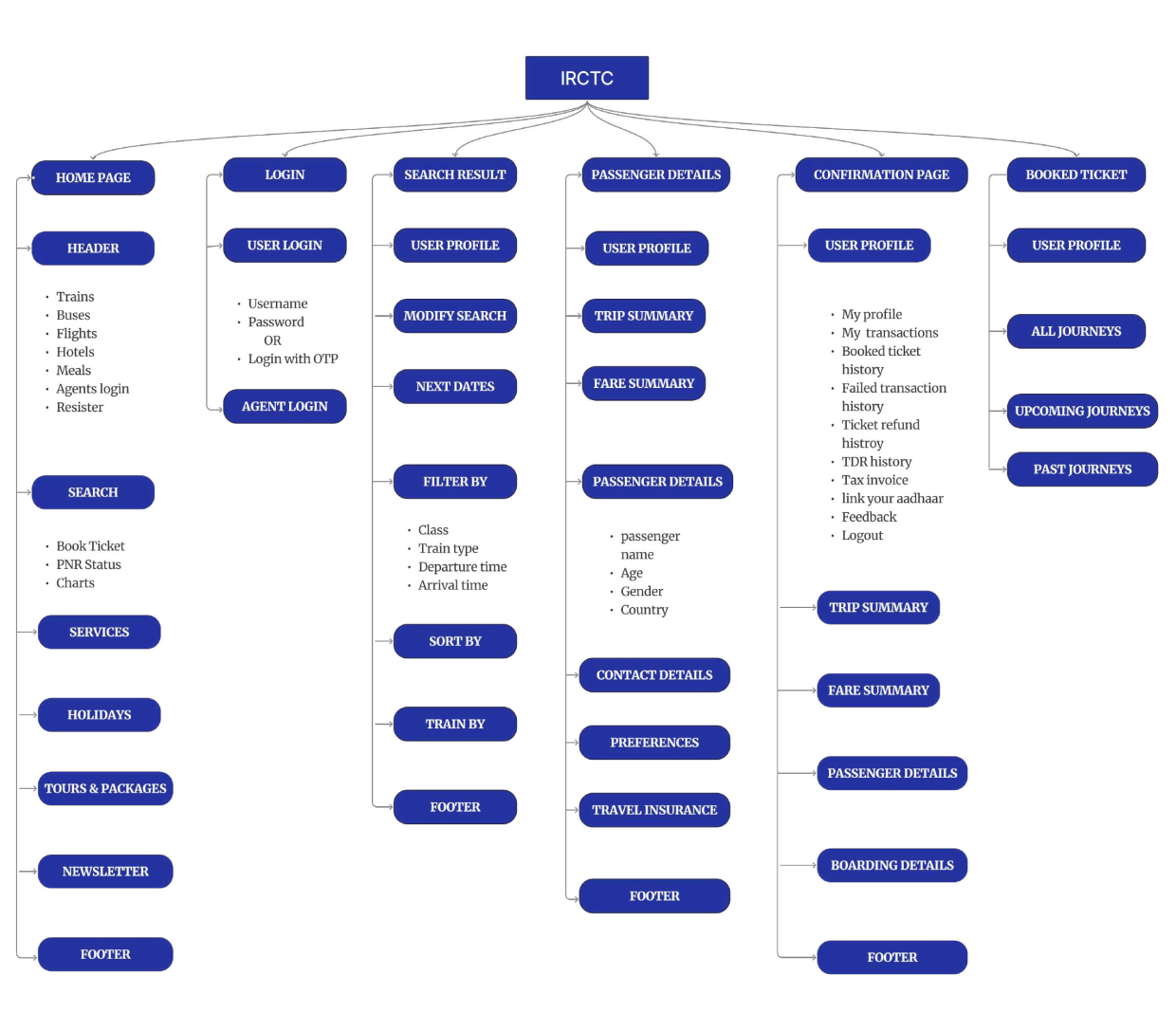
I Created A Journey Map To Visualize The Whole Process Of Users Completing Their Tasks, To Know What Pain Points They Are Facing At Each Step, And To See How I Can Improve Their Experience

Persona	Riya R.V			
Goal	Trying To Book A Ticket Through The IRCTC Website.			
Actions	Searching Trains	Login Process	Passenger Details	Downloading Ticket
Moods				
Task List	Searching For Trains. Applying Filters Choose The Perfect One From The List	Enter Username And Passwords. Entre Captcha Code Then Click Sign In.	Enter Passenger Details. Select Payment Mode Then Click Continue.	Try To Download Ticket. Looking For Print/Download Option.
Pain Points	Confused By Cluttered Design And Bad Hierarchy. Not Showing Tickets Without Login. Scroll All The Way Down To See Filters Not Showing Availability Of Seats Without Click On Refresh	Frustrated By The Advertisement That Is In The Middle Of The Login Page. Confused Between The Advertisement And The Captcha Because The Advertisement Is Stuck To The Captcha	No Profile Is Saved He Has To Enter Passenger Details Again No Option To Select A Preferred Seat Or Berth. Frustrated By Captchas On Every Single Page	It's Hard To Notice The Small. Print Icon On The Ticket
Opportunities	Clean And Minimal Design. There Is No Need To Log In To View Tickets.. Filters On Top So That Users Can See Them Without Scrolling The Availability Of Seats Should Be Visible Without Clicking On "Refresh	The Captcha Should Be Simple And Separated From The Advertisement. Login Should Be Clear And Simple	Login Should Be Clear And Simple. Provide A Preference Option For Seat And Berth Selection. The Captcha On The Login Page Is Enough	Actions For Tickets Should Be Clearly Visible



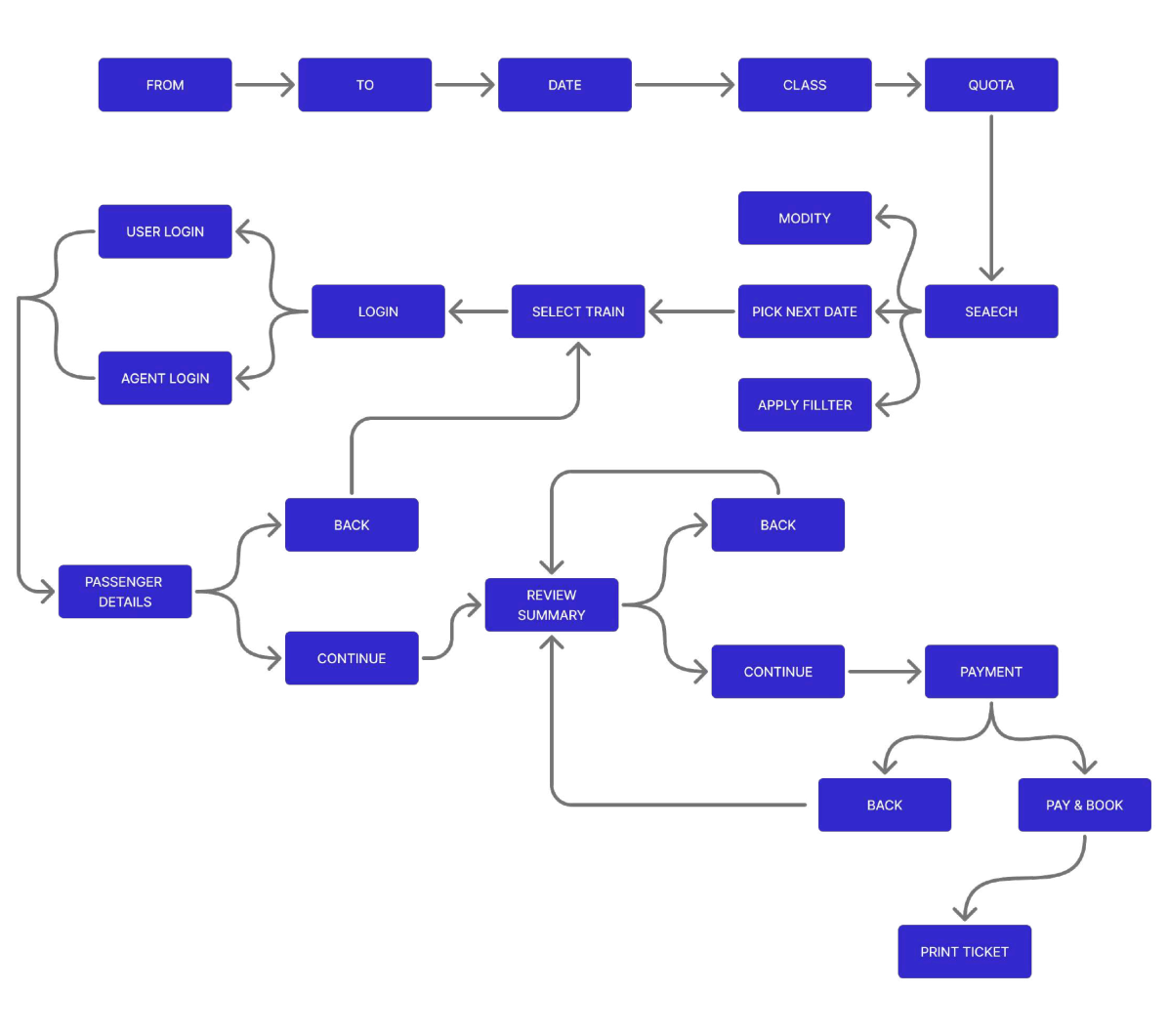
Information Architecture

To organize the date and determine what should be on which page i created an information architecture so that users could find all the important information easily .


IRCTC
HOME PAGE
HEADER
Trains
Buses
Flights
Hotels
Meals
Agents login
Resister
SEARCH
Book Ticket
PNR Status
Charts
SERVICES
HOLIDAYS
TOURS & PACKAGES
NEWSLETTER
FOOTER
LOGIN
USER LOGIN
Username
Password
OR
Login with OTP
AGENT LOGIN
SEARCH RESULT
USER PROFILE
MODIFY SEARCH
NEXT DATES
FILTER BY
Class
Train type
Departure time
Arrival time
SORT BY
TRAIN BY
FOOTER
PASSENGER DETAILS
USER PROFILE
TRIP SUMMARY
FARE SUMMARY
PASSENGER DETAILS
passenger name
Age
Gender
Country
CONTACT DETAILS
PREFERENCES
TRAVEL INSURANCE
FOOTER
CONFIRMATION PAGE
USER PROFILE
My profile
My transactions
Booked ticket history
Failed transaction history
Ticket refund history
TDR history
Tax invoice
link your aadhaar
Feedback
Logout
TRIP SUMMARY
FARE SUMMARY
PASSENGER DETAILS
BOARDING DETAILS
FOOTER
BOOKED TICKET
USER PROFILE
ALL JOURNEYS
UPCOMING JOURNEYS
PAST JOURNEYS

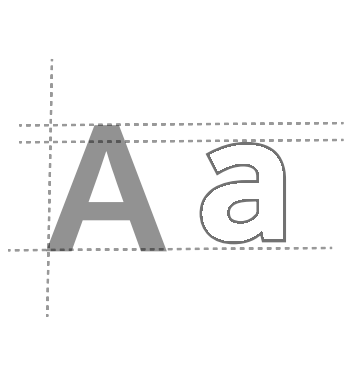
User Flow

Below is the main ticket booking flow of the website try to design a flow that is single and containt all the necessary steps with as few steps as possible.


FROM
TO
DATE
CLASS
QUOTA
MODIFY
SEAECH
PICK NEXT DATE
APPLY FILLTER
SELECT TRAIN
LOGIN
USER LOGIN
AGENT LOGIN
PASSENGER DETAILS
BACK
CONTINUE
REVIEW SUMMARY
BACK
CONTINUE
PAYMENT
BACK
PAY & BOOK
PRINT TICKET

Typography

Contemporary,
Adventurous,
Friendly & Versatile Font V



Semi Bold
Mukta
Regular
Medium

Color Palette

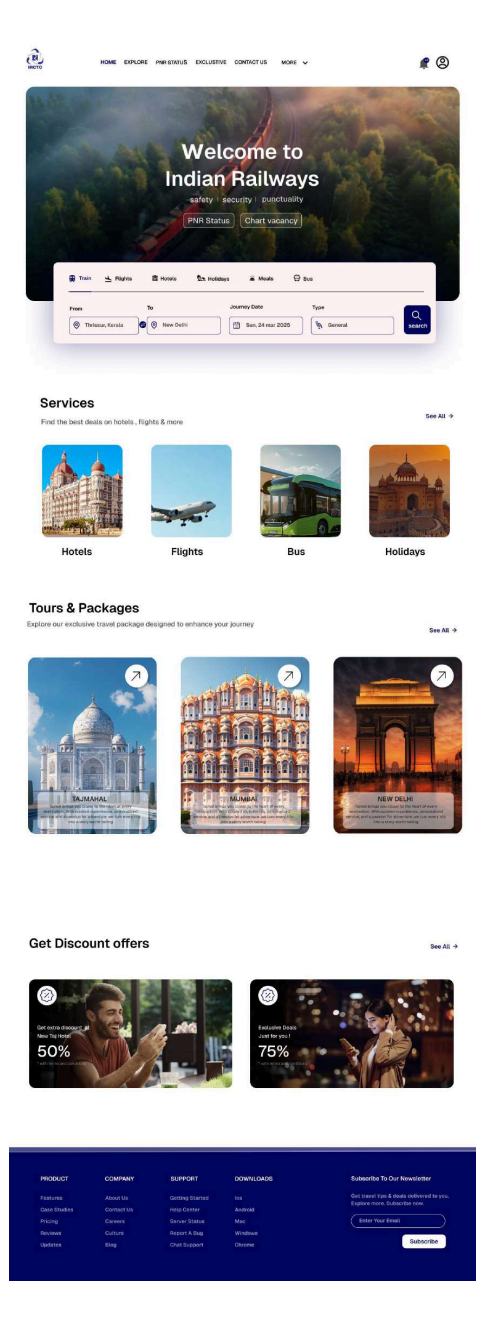
#03045E
PRIMARY

#F3F5F9
PRIMARY

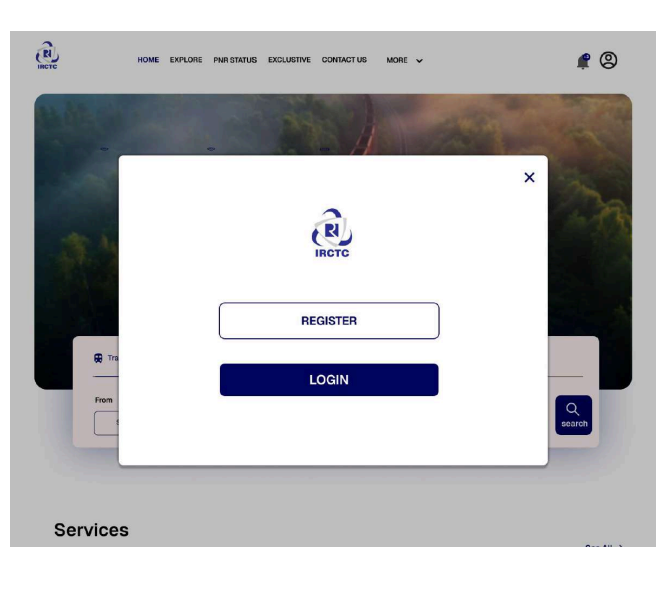
#313236 #2C2C78 #F3F5F9

Final Designs

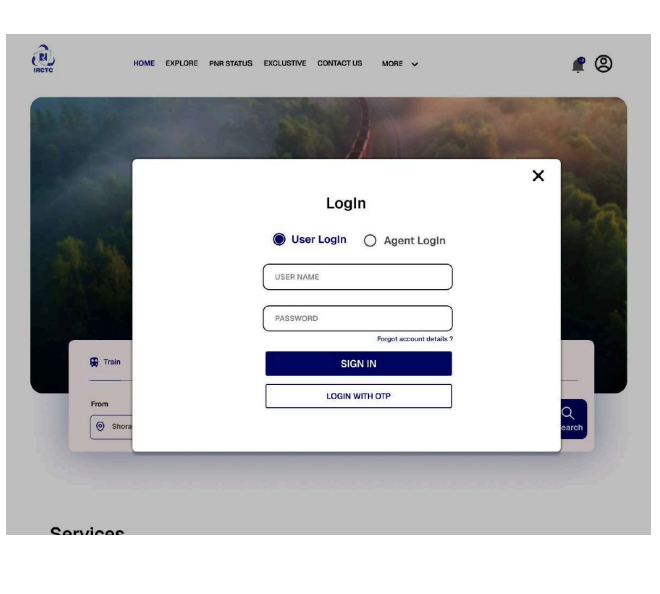
Home Page


Welcome to Indian Railways
Services
Hotels
Flights
Bus
Holidays
Tours & Packages
Get Discount offers
50%
75%

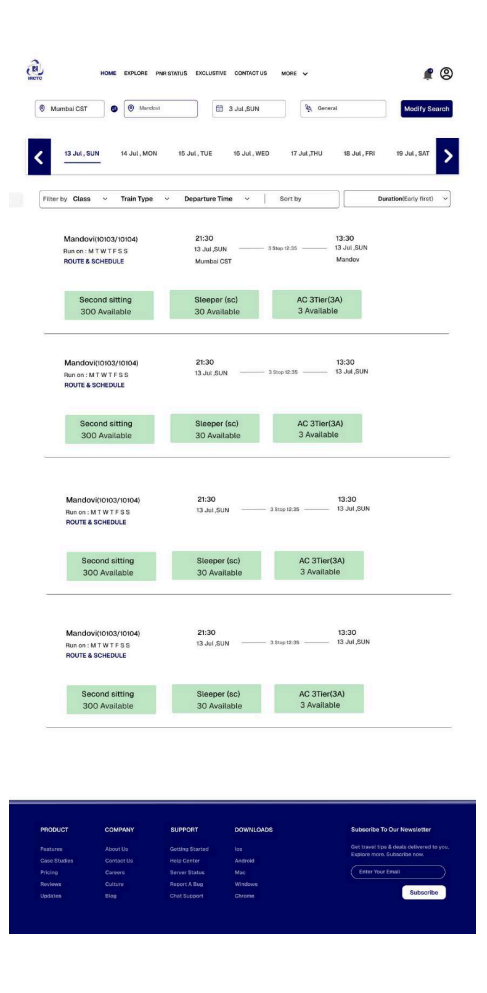
Sign Page


REGISTER
LOGIN
Services

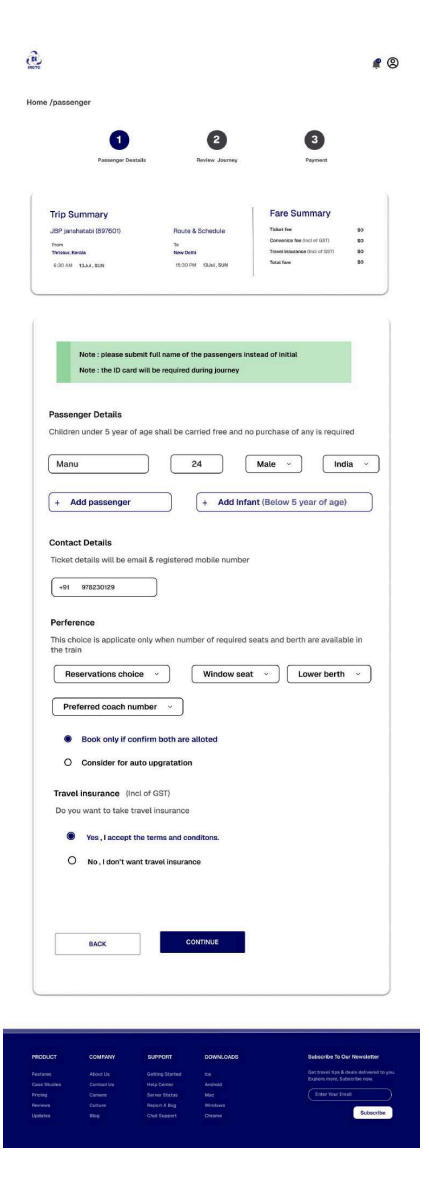
Login Page


Login
User Login
Agent Login
USER NAME
PASSWORD
SIGN IN
LOGIN WITH OTP

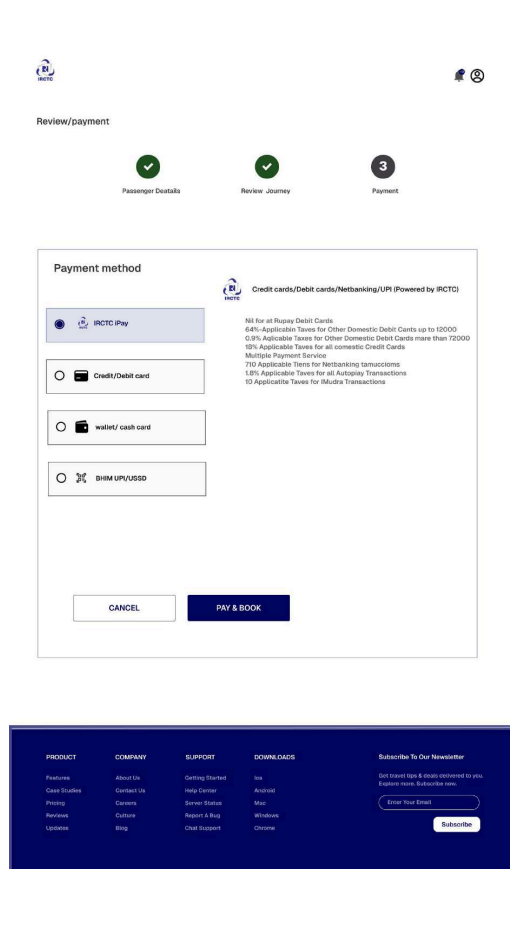
Train Listing / Search Result


Mumbai CST
Modify Search
Second sitting 300 Available
Sleeper (sc) 30 Available
AC 3Tier(3A) 3 Available

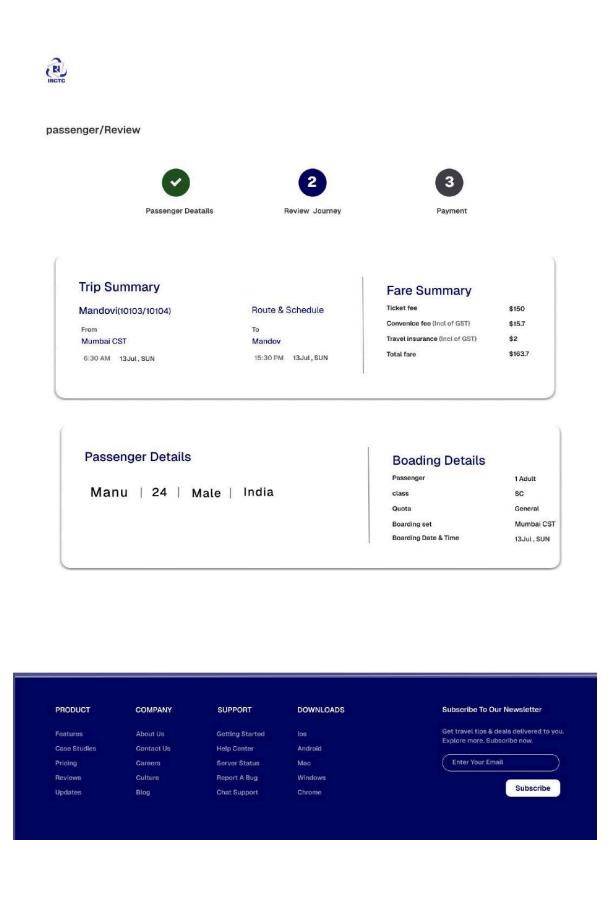
Passenger Details


Trip Summary
Fare Summary
Passenger Details
Contact Details
Perference
Travel insurance
BACK
CONTINUE

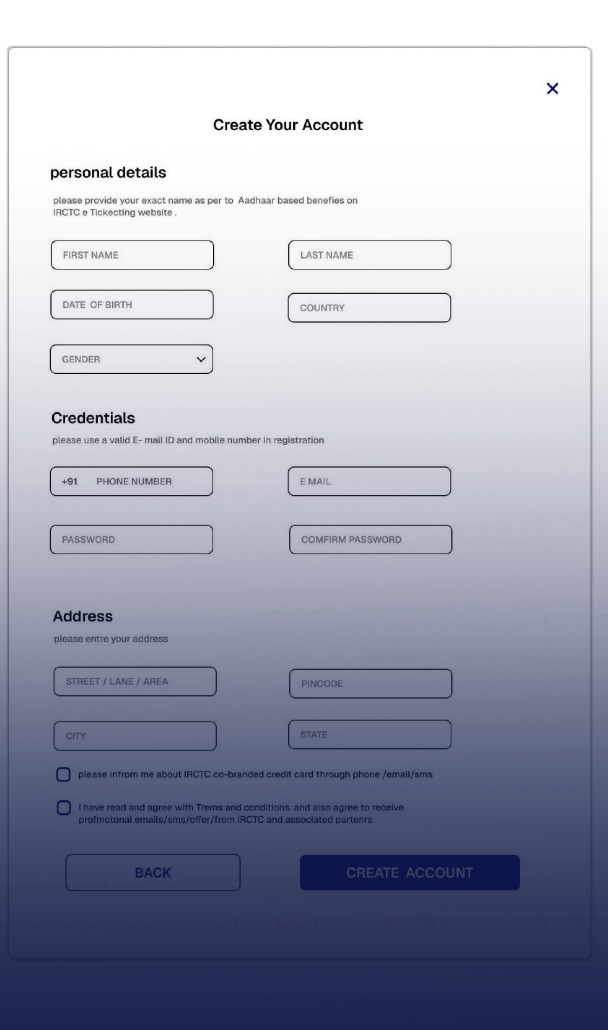
Payment Page


Review/payment
Payment method
CANCEL
PAY & BOOK

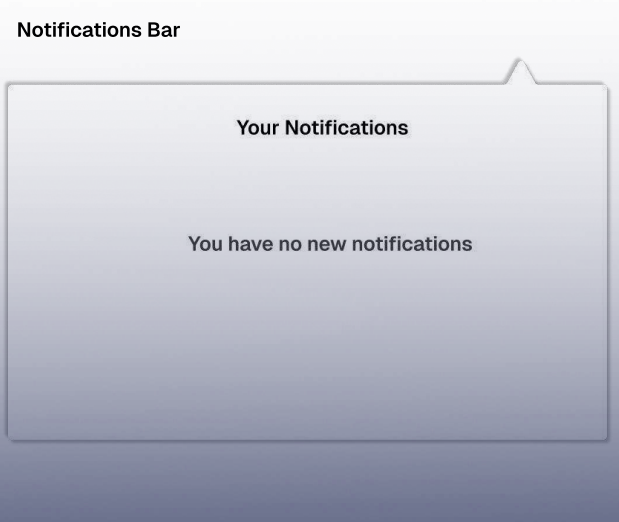
Comfirmation Page


passenger/Review
Trip Summary
Fare Summary
Passenger Details
Manu | 24 | Male | India
Boarding Details

Create Account Page


Create Your Account
personal details
FIRST NAME
LAST NAME
DATE OF BIRTH
COUNTRY
GENDER
Credentials
+91 PHONE NUMBER
E-MAIL
PASSWORD
COMFIRM PASSWORD
Address
STREET / LANE / AREA
PINCODE
CITY
STATE
BACK
CREATE ACCOUNT

Notifications Bar



Your Notifications

You have no new notifications

THANK YOU!

Feel free to share feedback and appreciation!

im.saileshh@gmail.com